Physics 9702 Paper 4 · Topic 22

Quantum physics

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/42 February/March 2022, Question 8 (7 marks) page 2

Examiner report: (ii) Many candidates realised the wave behaviour of electrons was being shown, but very few added that this was only true for moving electrons. (iii) Only the strongest candidates were able to achieve credit for relating the interatomic spacing in the crystal to the de Broglie wavelength of the electron beam.

2 9702/42 February/March 2023, Question 7 (7 marks) page 4

Examiner report: (b) Candidates generally struggled with this question. However, only the strongest candidates were able to correctly process the energy change, realising that the sign of the answer must be negative, and that the data requires a 3 significant figure answer.

3 9702/41 October/November 2022, Question 8 (10 marks) page 6

Examiner report: (c) These two explanations were difficult.

4 9702/41 October/November 2023, Question 8 (10 marks) page 7

Examiner report: (c) Very few candidates realised that the photons would have greater momentum but there would also be fewer photons per unit time and hence the pressure would be unchanged.

5 9702/42 February/March 2025, Question 8 (10 marks) page 9

Examiner report: (iii) This was a challenging question.

6 9702/42 October/November 2020, Question 11 (10 marks) page 10

Examiner report: For candidates who did give responses that answered the question, there was often confusion between photons and electrons (with some candidates discussing the frequency of incident electrons, and others discussing the speed of emitted photons). Only the strongest candidates correctly calculated the threshold frequency.

7 9702/42 February/March 2020, Question 10 (7 marks) page 12

Examiner report: (c) (i) This question proved to be very challenging.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

8 **(a)** State the formula for the de Broglie wavelength λ of a moving particle.

State the meaning of any other symbol used.

.....

.....

..... [2]

(b) Electrons accelerate through a potential difference, pass through a thin crystal and are then incident on a fluorescent screen.

The pattern in Fig. 8.1 is observed on the fluorescent screen.

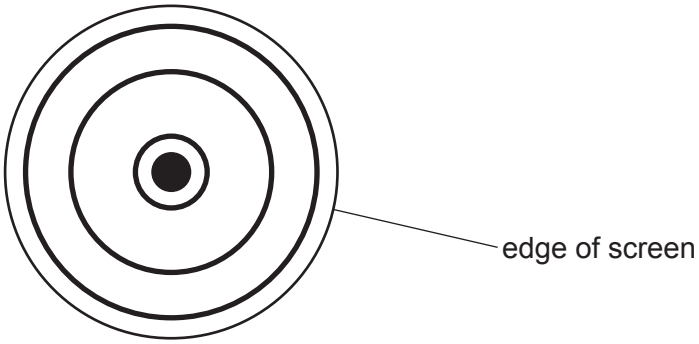


Fig. 8.1 not to scale

(i) State the name of the phenomenon shown by the electrons at the crystal.

..... [1]

(ii) State what this phenomenon shows about the nature of electrons.

.....

..... [1]

(iii) Suggest why the thin crystal causes the phenomenon in **(b)(i)**.

.....

..... [1]

(iv) The electron is accelerated through a different potential difference. The new pattern observed on the screen is shown in Fig. 8.2.

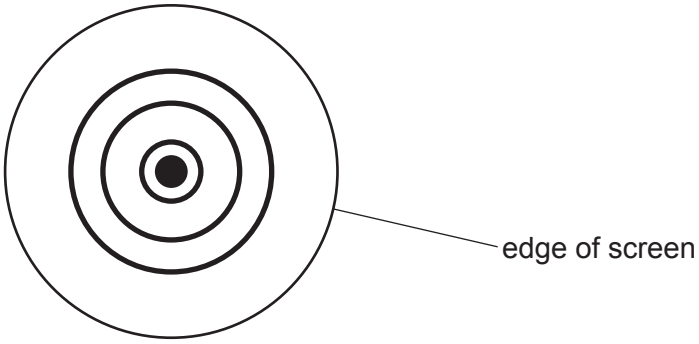


Fig. 8.2 not to scale

State and explain the change that has been made to the potential difference to create the pattern shown in Fig. 8.2.

.....

.....

.....

..... [2]

[Total: 7]

7 **(a)** A beam of white light passes through a cloud of cool gas. The spectrum of the transmitted light is viewed and contains a number of dark lines.

Explain why these dark lines occur.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Some energy levels for the electron in an isolated hydrogen atom are illustrated in Fig. 7.1.

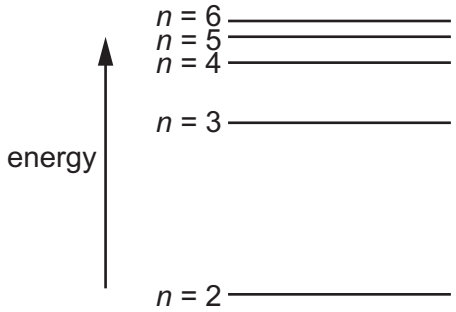


Fig. 7.1

Table 7.1 shows the wavelengths of photons that are emitted in the transitions to $n = 2$ from the other energy levels shown in Fig. 7.1.

Table 7.1

wavelength / nm
412
435
488
658

The energy associated with the energy level $n = 2$ is $-3.40\,\text{eV}$.

Calculate the energy, in J, of energy level $n = 3$.

energy = ... J [3]

[Total: 7]

8 **(a)** State what is meant by the work function energy of a metal.

.....

.....

..... [2]

(b) Ultraviolet radiation of frequency 1.36×10^{15} Hz is incident, in a vacuum, on a metal surface. The power of the radiation incident on the surface is 8.36 mW. Photoelectrons are emitted with a maximum kinetic energy of 3.09×10^{-19} J.

(i) Determine the number of photons incident on the surface per unit time.

number per unit time = ... s^{-1} [2]

(ii) Calculate the work function energy Φ of the metal.

Φ = ... J [2]

(c) The frequency of the radiation incident on the surface in **(b)** is increased while the power remains constant.

State and explain the effect of this change on:

(i) the maximum kinetic energy of the photoelectrons

.....

.....

..... [2]

(ii) the rate of emission of photoelectrons.

.....

.....

..... [2]

[Total: 10]

8 **(a)** **(i)** Show that the momentum p of a photon of electromagnetic radiation with wavelength λ is given by

$$p = \frac{h}{\lambda}$$

where h is the Planck constant.

[2]

(ii) Use the expression in **(a)(i)** to show that a photon in free space that has a momentum of 9.5×10^{-28} N s is a photon of red light.

[1]

(b) A beam of red light of intensity 160 W m^{-2} is incident normally on a plane mirror, as shown in Fig. 8.1. The momentum of each photon in the beam is 9.5×10^{-28} N s.

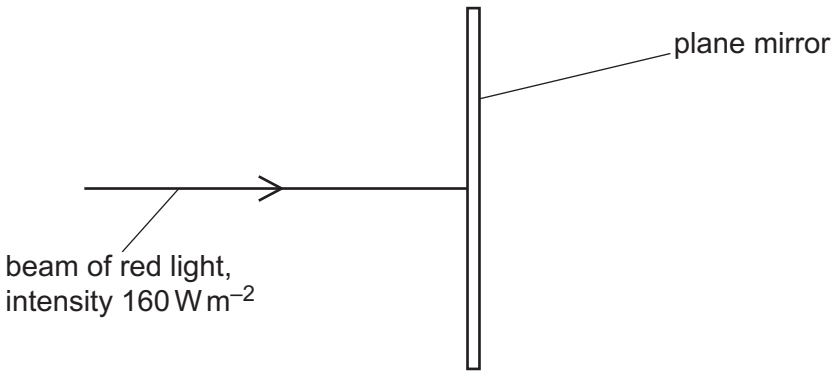


Fig. 8.1

All of the light is reflected by the mirror in the opposite direction to its original path.
The cross-sectional area of the beam is 2.5×10^{-6} m^2.

(i) Show that the number of photons incident on the mirror per unit time is $1.4 \times 10^{15}\,s^{-1}$.

[2]

(ii) Use the information in **(b)(i)** to determine the pressure exerted by the light beam on the mirror.

pressure = ... Pa [3]

(c) The beam of red light in **(b)** is now replaced with a beam of blue light of the same intensity.

Suggest and explain whether the pressure exerted on the mirror by the beam of blue light is less than, the same as, or greater than the pressure exerted by the beam of red light.

.....

.....

.....

..... [2]

[Total: 10]

8 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) A laser emits red light of a single wavelength. The light is produced when electrons move from a higher energy level to a lower energy level. The difference in energy between the two levels is 1.96 eV.

(i) Calculate the wavelength of the light.

wavelength = ... m [3]

(ii) The power of the beam emitted by the laser is 1.0×10^{-2} W.

Calculate the number of photons emitted per unit time by the laser.

number per unit time = ... s^{-1} [1]

(iii) The photons are incident normally on a surface. Half of the number of photons are absorbed by the surface, and half are reflected.

Determine the average force exerted by the beam of photons on the surface.

average force = ... N [4]

[Total: 10]

11 **(a)** Electromagnetic radiation is incident on a metal surface.

It is observed that there is a minimum frequency of electromagnetic radiation below which emission of electrons does not occur.

This observation provides evidence for a particulate nature of electromagnetic radiation.

State **two** other observations associated with photoelectric emission that provide evidence for a particulate nature of electromagnetic radiation.

1. .....

.....

2. .....

.....

[2]

(b) The maximum kinetic energy E_{MAX} of electrons emitted from a metal surface is determined for different wavelengths λ of the electromagnetic radiation incident on the surface.

The variation with $\frac{1}{\lambda}$ of E_{MAX} is shown in Fig. 11.1.

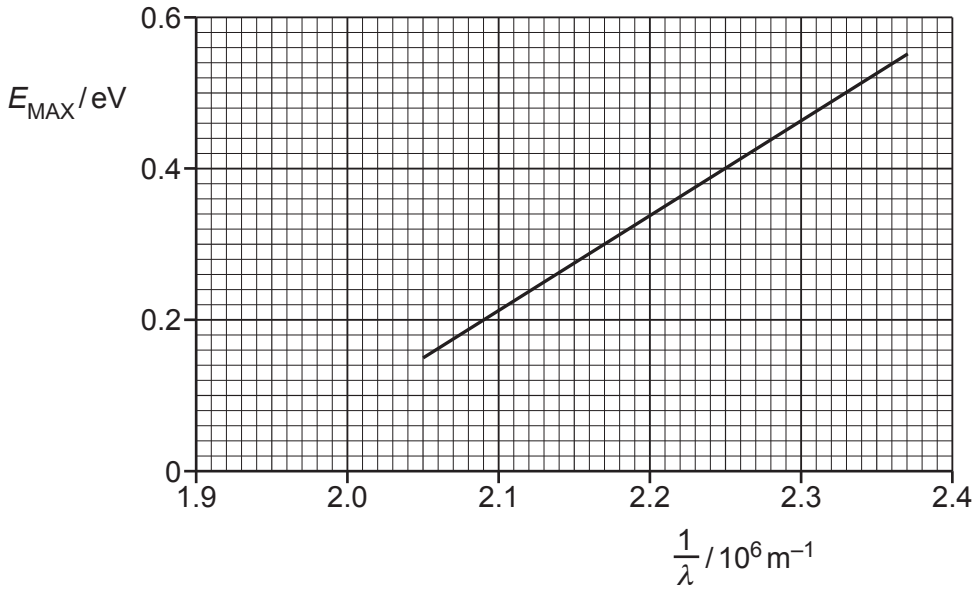


Fig. 11.1

(i) Use Fig. 11.1 to determine the threshold frequency f_0.

f_0 = ... Hz [2]

(ii) Use the gradient of the line on Fig. 11.1 to determine a value for the Planck constant h. Explain your working.

h = ...J s [4]

(c) The electromagnetic radiation is now incident on a metal with a larger work function energy than the metal in **(b)**.

On Fig. 11.1, sketch the variation with $\frac{1}{\lambda}$ of E_{MAX}. [2]

[Total: 10]

10 **(a)** By reference to the photoelectric effect, explain what is meant by *work function energy*.

.....

.....

..... [2]

(b) In an experiment, electromagnetic radiation of frequency f is incident on a metal surface.

The results in Fig. 10.1 show the variation with frequency f of the maximum kinetic energy E_{MAX} of electrons emitted from the surface.

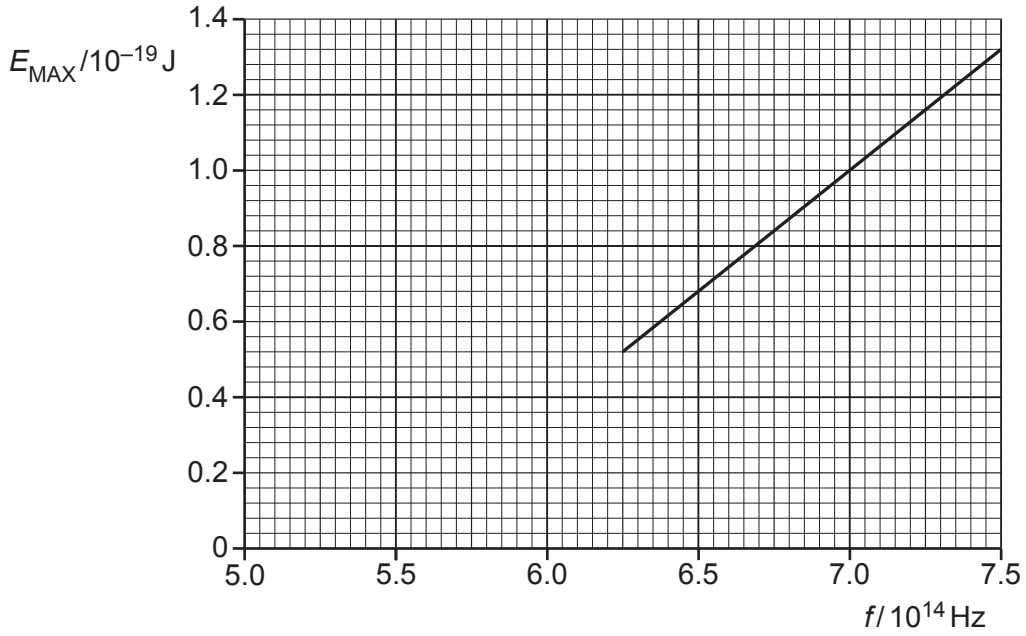


Fig. 10.1

(i) Determine the work function energy in J of the metal used in the experiment.

work function energy = ... J [2]

(ii) The work function energy in eV for some metals is given in Table 10.1.

Table 10.1

metal	work function / eV
tungsten	4.49
magnesium	3.68
potassium	2.26

Determine the metal used in the experiment. Show your working.

.....

..... [1]

(c) The intensity of the electromagnetic radiation for one particular frequency in **(b)** is increased.

State and explain the change, if any, in:

(i) the maximum kinetic energy of the emitted electrons

.....

..... [1]

(ii) the rate of emission of photoelectrons.

.....

..... [1]

[Total: 7]

Question	Answer	Marks
8(a)	$\lambda = \frac{h}{p}$ or $\lambda = \frac{h}{mv}$	**M1**
	where *h* is the Planck constant **and** *p* is the momentum (of particle) / *mv* is the momentum (of particle) / *m* is the mass (of particle) and *v* is the velocity (of particle)	**A1**
8(b)(i)	(electron) diffraction	**B1**
8(b)(ii)	moving electrons behave like waves	**B1**
8(b)(iii)	spacing between atoms ≈ wavelength of electron *or* diameter of atom ≈ wavelength of electron	**B1**
8(b)(iv)	Any one of: • wavelength has decreased • electron had greater momentum	**M1**
	so (accelerating) p.d. was increased	**A1**

Question	Answer	Marks
7(a)	photon absorbed (by electron) and electron excited	**B1**
	photon energy equal to difference in (energy of two) energy levels	**B1**
	photon energy relates to a single wavelength / single frequency	**B1**
	electron de-excites and emits photon in any direction	**B1**
7(b)	$\frac{hc}{\lambda} = \Delta E$	**C1**
	uses 658 nm	**C1**
	$\frac{6.63\times10^{-34}\times3.00\times10^{8}}{658\times10^{-9}} = -E_1 - (-3.40 \times 1.60 \times 10^{-19})$ $E_1 = -2.42 \times 10^{-19}$ J	**A1**

Question	Answer	Marks
8(a)	photon energy (to remove electron)	**B1**
	minimum energy to remove electron **or** energy to remove electron from surface **or** energy to remove electron with zero kinetic energy	**B1**
8(b)(i)	photon energy = hf	**C1**
	number per unit time = $8.36 \times 10^{-3} / (1.36 \times 10^{15} \times 6.63 \times 10^{-34})$ $= 9.27 \times 10^{15}\ s^{-1}$	**A1**
8(b)(ii)	$hf = \Phi + E_{MAX}$	**C1**
	$\Phi = (1.36 \times 10^{15} \times 6.63 \times 10^{-34}) - (3.09 \times 10^{-19})$ $= 5.93 \times 10^{-19}$ J	**A1**
8(c)(i)	greater photon energy (and same work function)	**M1**
	so maximum kinetic energy is increased	**A1**
8(c)(ii)	(greater photon energy and same power so) lower number of photons (per unit time)	**M1**
	(each electron absorbs one photon) so lower rate of emission	**A1**

Question	Answer	Marks
8(a)(i)	$p = E / c$	**M1**
	$E = hc / \lambda$ and completion of algebra leading to $p = h / \lambda$	**A1**
8(a)(ii)	wavelength = $(6.63 \times 10^{-34}) / (9.5 \times 10^{-28}) = 700 \times 10^{-9}$ m so red	**B1**
8(b)(i)	power = intensity × area	**C1**
	number per unit time = $(160 \times 2.5 \times 10^{-6}) / (9.5 \times 10^{-28} \times 3.00 \times 10^{8}) = 1.4 \times 10^{15}$ s^{-1}	**A1**
8(b)(ii)	pressure = force / area	**C1**
	force = rate of change of momentum $= 2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}$	**C1**
	pressure = $(2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}) / (2.5 \times 10^{-6})$ $= 1.1 \times 10^{-6}$ Pa	**A1**
8(c)	photons have greater momentum **or** fewer photons per unit time	**B1**
	greater photon momentum but smaller number of photons (per unit time) so pressure is the same	**B1**

Question	Answer	Marks
8(a)	• quantum of energy	M1
	• of electromagnetic radiation	A1
8(b)(i)	$(\Delta)E = hc / \lambda$	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (1.96 \times 1.60 \times 10^{-19})$	C1
	$= 6.3 \times 10^{-7}$ m	A1
8(b)(ii)	number per unit time = power / energy per photon $= (1.0 \times 10^{-2}) / (1.96 \times 1.60 \times 10^{-19})$ $= 3.2 \times 10^{16}$ s^{-1}	A1

Question	Answer	Marks
8(b)(iii)	*either*: force = rate of change of momentum *or*: $F = \Delta p / t$	C1
	$p = E / c$	C1
	half the photons have change in momentum p, the other half have change in momentum $2p$	C1
	$F = [(1.96 \times 1.60 \times 10^{-19}) / (3.00 \times 10^{8})] \times 3.2 \times 10^{16} \times [(2 + 1) / 2]$ $= 5.0 \times 10^{-11}$ N	A1

Question	Answer	Marks
11(a)	any two points from: • (maximum) kinetic energy of electrons is independent of intensity • maximum kinetic energy of electrons depends on frequency • no time delay (between illumination and emission)	**B2**
11(b)(i)	(for E_{MAX} = 0,) $1 / \lambda_0 = 1.93 \times 10^6$ (m^{-1})	**C1**
	$f_0 = 3.00 \times 10^8 \times 1.93 \times 10^6$ $= 5.8 \times 10^{14}$ Hz	**A1**
11(b)(ii)	$hc / \lambda = \Phi + E_{MAX}$	**C1**
	hc = gradient	**C1**
	gradient = e.g. $[(0.40 – 0.20) \times 1.60 \times 10^{–19}] / [(2.25 – 2.09) \times 10^6]$ (*working needed*) $(= 2.0 \times 10^{–25})$	**M1**
	$h = (2.0 \times 10^{–25}) / (3.00 \times 10^8) = 6.7 \times 10^{–34}$ J s (*both working and answer needed*)	**A1**
11(c)	straight line with same gradient as the original	**B1**
	straight line with x-axis intercept greater than 1.93×10^6 $m^{–1}$	**B1**

Question	Answer	Marks
10(a)	energy of a photon required to remove an electron	**B1**
	either: energy to remove electron from a surface *or*: minimum energy to remove electron *or*: energy to remove electron with zero kinetic energy	**B1**
10(b)(i)	Correct read off from graph of f as 5.45×10^{14} Hz when $E_{MAX} = 0$ $5.45 \times 10^{14} \times 6.63 \times 10^{-34}$	**C1**
	= 3.6×10^{-19} J	**A1**
10(b)(ii)	$3.6 \times 10^{-19} / 1.6 \times 10^{-19} = 2.3$ eV so potassium	**A1**
10(c)(i)	each photon has same energy so no change	**B1**
10(c)(ii)	more photons (per unit time) so (rate of emission) increases	**B1**